MINISTRY OF EDUCATION AND TRAINING MINISTRY OF HEALTH

HANOI MEDICAL UNIVERSITY

PHAM HOANG ANH

HPV VACCINATION UPTAKE AND ASSOCIATED FACTORS AMONG VIETNAMESE WOMEN AGED 15–29 YEARS: A SECONDARY DATA ANALYSIS OF THE 2020–2021 MICS 6 SURVEY

MEDICAL DOCTOR GRADUATION THESIS

COURSE 2020 - 2026

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Institute : School of Preventive Medicine and Public Health
Major : Preventive Medicine

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Scientific supervisors

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HANOI – 2026

ACKNOWLEDGEMENTS

First and foremost, I would like to express my deepest gratitude and sincerest thanks to my supervisors, Assoc. Prof. Pham Bich Diep, PhD, and Pham Thanh Tung, PhD, for their invaluable guidance, dedicated support, and profound expertise throughout the process of conducting this graduation thesis. Their insightful mentorship and constant encouragement have been instrumental in the completion and quality of this research.

I would also like to extend my heartfelt appreciation to the MICS 6 (Multiple Indicator Cluster Survey Round 6) from UNICEF team for providing the necessary dataset and granting permission for its use in this study. This research would not have been possible without the high-quality data and resources they provided.

My sincere thanks go to the lecturers and staff of the School of Preventive Medicine and Public Health, Hanoi Medical University, for providing me with a solid academic foundation and essential methodological skills during my years of study. The knowledge gained from the program has played a crucial role in my academic development.

To my classmates and friends—thank you for the spirited discussions and shared journey through medical school.

Above all, I owe an immeasurable debt to my family for their unconditional love, patience, and sacrifice. You are my greatest motivation

Hanoi, 2026

PHAM HOANG ANH

ABSTRACT

Background. Despite the proven efficacy of HPV vaccination, substantial disparities in awareness and uptake persist in low- and middle-income countries. In Vietnam, national evidence on the magnitude and determinants of these inequalities among young women remains limited.

Objectives. This study aimed to (1) estimate the weighted prevalence of HPV vaccine awareness and vaccination among Vietnamese women aged 15–29; (2) quantify socioeconomic inequalities using the concentration index; and (3) decompose the observed inequality via the Erreygers method.

Methods. A cross-sectional analysis of the Vietnam Multiple Indicator Cluster Survey 2020–2021 (MICS6) was conducted. The analytic sample comprised 4,102 women aged 15–29. Survey-weighted descriptive statistics, bivariate chi-squared tests, Poisson regression yielding prevalence ratios (PR), the Wagstaff concentration index, and Erreygers decomposition were employed.

Results. Overall, 62.4% of women reported awareness of the HPV vaccine, but only 7.5% had been vaccinated. A significant pro-rich inequality was observed in both awareness (Erreygers CI = 0.371) and vaccination uptake (Erreygers CI = 0.116). The awareness–vaccination gap was concentrated among lower socioeconomic groups, with wealth and education contributing most to this inequality pattern.

Conclusion. A substantial awareness–vaccination gap was observed among Vietnamese women aged 15–29 years. Socioeconomic inequalities in both awareness and vaccination uptake were evident, with wealth and education contributing substantially to the measured inequality. As Vietnam moves toward integrating HPV vaccination into the Expanded Program on Immunization, these findings highlight the need for equity-oriented strategies that address affordability, information access, and support for disadvantaged groups.

Keywords: HPV vaccination; socioeconomic inequality; concentration index; Erreygers decomposition; MICS; Vietnam.

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LIST OF ABBREVIATIONS

Abbreviation	Full term
CI	Concentration Index
CIN	Cervical Intraepithelial Neoplasia
DHS	Demographic and Health Survey
ECI	Erreygers-Corrected Concentration Index
EPI	Expanded Programme on Immunization
GAVI	Global Alliance for Vaccines and Immunisation
HPV	Human Papillomavirus
ICT	Information and Communication Technology
IRB	Institutional Review Board
LMIC	Low- and Middle-Income Country
MM	Mass Media
MICS	Multiple Indicator Cluster Survey
OR	Odds Ratio
PR	Prevalence Ratio
PSU	Primary Sampling Unit
SES	Socioeconomic Status

Abbreviation	Full term
UNICEF	United Nations Children’s Fund
VIA	Visual Inspection with Acetic Acid
WHO	World Health Organization

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INTRODUCTION

Cervical cancer (CC) is currently one of the most significant public health challenges globally; it is ranked as the 8th most common cancer by incidence and the 9th leading cause of cancer-related mortality. The primary cause of this disease is the Human Papillomavirus (HPV). The association between certain oncogenic (high-risk) strains of HPV and cervical cancer is well established. Although HPV is essential to the transformation of cervical epithelial cells, it is not sufficient, and a variety of cofactors and molecular events influence whether cervical cancer will develop. According to the U.S. Centers for Disease Control and Prevention (CDC), approximately 80% of adults will contract HPV at least once in their lifetime. Consequently, HPV vaccination is regarded as a pivotal advancement in modern medicine, effectively preventing HPV-related diseases, especially cervical cancer.

Reports from the World Health Organization (WHO) in 2019 estimated approximately 620,000 new cancer cases in women and 70,000 in men linked to HPV. Notably, in 2022 alone, the world recorded about 660,000 new cases of cervical cancer, resulting in 350,000 deaths. A study of over one million women across five continents indicated a global HPV prevalence of 11.7%. The highest prevalence was found in Sub-Saharan Africa (24%), followed by Latin America and the Caribbean (16%), Eastern Europe (14%), and Southeast Asia (14%).

In Vietnam, according to the 2021 Sustainable Development Goals (SDGs) Indicators Report on Children and Women by the General Statistics Office, among women and girls aged 15–29 who had heard of the HPV vaccine, only 12% had been vaccinated against HPV. , and only 28% of women aged 30–49 had ever undergone cervical cancer screening. These figures reveal a severe deficit in proactive prevention, posing significant challenges for both national and global healthcare systems. In response, the World Health Assembly (WHA) adopted a global strategy to eliminate cervical cancer as a public health problem, with three specific targets: 90% of girls fully vaccinated against HPV by age 15; 70% of women screened using a high-performance test at ages 35 and 45; and 90% of women diagnosed with cervical disease receiving treatment. In Vietnam, while the HPV vaccine has been available through private services since 2008, the Ministry of Health’s "National Action Plan for Cervical Cancer Prevention and Control 2016–2025" set a target of at least 25% of girls and women being vaccinated by 2025.

Although the primary target group for HPV vaccination is girls aged 9–14 years, women aged 15–29 remain relevant for evaluating catch-up vaccination needs and socioeconomic barriers to uptake in Vietnam. Recent studies have highlighted various factors influencing vaccination behavior, with socioeconomic status being a prominent driver. Globally, HPV vaccine coverage is only about 39.7%, with stark disparities across income groups.

Furthermore, recent findings from 2025 indicate that HPV screening and vaccination uptake in Southern Vietnam remains alarmingly low. Only 4% of Southern Vietnamese women were found to be vaccinated, and only 40% had ever undergone screening - figures far below the WHO’s 2030 targets. This regional discrepancy highlights the complex interplay between geographic location, socioeconomic barriers, and health perceptions.

Despite rising cervical cancer rates, we still lack a complete picture of why HPV vaccination remains low in Vietnam. It's unclear how social factors combine to hinder widespread access. Addressing these obstacles is now a critical mission for the nation's healthcare leaders. We conducted this study using a decomposition analysis based on secondary data from the MICS 6 to assess the prevalence of HPV vaccination and its associated factors. The study focuses on two primary objectives:

1. Describe the weighted prevalence of HPV vaccine awareness and HPV vaccination among women aged 15-29 in Vietnam, stratified by sociodemographic characteristics.
2. Quantify the magnitude of socioeconomic inequality in HPV vaccine awareness and vaccination using the concentration index.
3. Identify and quantify the contribution of socioeconomic determinants to the observed inequality through Erreyger's decomposition analysis.

CHAPTER I: LITERATURE REVIEW

1.1. Overview of Cervical cancer and prevention

1.1.1. Definition of cervical cancer

Cervical cancer develops in a woman's cervix (the entrance to the uterus from the vagina). Cervical cancer is cancer that starts in the cells of the cervix. The cervix is the lower, narrow end of the uterus (womb). The cervix connects the uterus to the vagina (birth canal). Cervical cancer usually develops slowly over time. Before cancer appears in the cervix, the cells of the cervix go through changes known as dysplasia, in which abnormal cells begin to appear in the cervical tissue. Over time, if not destroyed or removed, the abnormal cells may become cancer cells and start to grow and spread more deeply into the cervix and to surrounding areas. Cervical cancers are named after the type of cell where the cancer started. The two main types are:

Squamous cell carcinoma: Most cervical cancers (up to 90%) are squamous cell carcinomas. These cancers develop from cells in the ectocervix.

Adenocarcinoma: Cervical adenocarcinomas develop in the glandular cells of the endocervix. Clear cell adenocarcinoma, also called clear cell carcinoma or mesonephroma, is a rare type of cervical adenocarcinoma.

Sometimes, cervical cancer has features of both squamous cell carcinoma and adenocarcinoma. This is called mixed carcinoma or adenosquamous carcinoma. Very rarely, cancer develops in other cells in the cervix.

1.1.2. The burden of cervical cancer

Cervical cancer is the fourth most common cancer in women. In 2022, an estimated 660 000 women were diagnosed with cervical cancer worldwide and about 350 000 women died from the disease. Cervical cancer is the most commonly diagnosed cancer in 23 countries and is the leading cause of cancer death in 36 countries. The vast majority of these countries are in sub-Saharan Africa, Melanesia, South America, and South-Eastern Asia

Effective primary (HPV vaccination) and secondary prevention approaches (screening for, and treating precancerous lesions) will prevent most cervical cancer cases. When diagnosed, cervical cancer is one of the most successfully treatable forms of cancer, as long as it is detected early and managed effectively. Cancers diagnosed in late stages can also be controlled with appropriate treatment and palliative care. With a comprehensive approach to prevent, screen and treat, cervical cancer can be eliminated as a public health problem within a generation. The targets of the global strategy are, by 2030:

▪ To vaccinate 90% of eligible girls against HPV;

▪ To screen 70% of eligible women at least twice in their lifetimes; and

▪ To effectively treat 90% of those with a positive screening test or a cervical lesion, including palliative care when needed

1.2 Overview of the Human Papillomavirus and HPV vaccination

1.2.1 Characteristics of HPV

Almost all cervical cancer cases (99%) are linked to infection with high-risk human papillomaviruses (HPV), an extremely common virus transmitted through sexual contact. Although most infections with HPV resolve spontaneously and cause no symptoms, persistent infection can cause cervical cancer in women.

Human papillomavirus (HPV) is a small, non-enveloped deoxyribonucleic acid (DNA) virus that infects skin or mucosal cells. The circular, double-stranded viral genome is approximately 8-kb in length. The genome encodes for 6 early proteins responsible for virus replication and 2 late proteins, L1 and L2, which are the viral structural proteins. At least 13 of more than 100 known HPV genotypes can cause cancer of the cervix and are associated with other anogenital cancers and cancers of the head and neck. The two most common "high-risk" genotypes (HPV 16 and 18) cause approximately 70% of all cervical cancers. HPV was estimated to cause almost half a million cases and 250,000 deaths from cervical cancer in 2002, of which about 80% occurred in developing countries. Two "low-risk" genotypes (HPV 6 and 11) cause genital warts, a common benign condition of the external genitalia that causes significant morbidity. HPV is highly transmissible, with peak incidence soon after the onset of sexual activity, and most persons acquire infection at some time in their lives.

1.2.2 Epidemiology of HPV

HPV subtypes show a predilection for body sites they most commonly infect, and disease manifestations that result from infection may vary. Over 180 subtypes of HPV have been identified. Cutaneous warts of the hands and feet, such as verruca vulgaris or verruca plantaris, are most commonly caused by HPV subtypes 1, 2, 4, 27, or 57. Most anogenital warts, such as condyloma acuminatum, are caused by HPV subtypes 6 or 11 and termed low-risk HPV; these subtypes also are responsible for juvenile and adult recurrent respiratory papillomatosis. Pre-cancerous and cancerous lesions of the cervix, male and female anogenital areas, and oropharyngeal area are most commonly caused by HPV subtypes 16 and 18. However, subtypes 31, 33, 35, 45, 52, and 58 also fall in the high-risk HPV group as they are associated with cervical cancer development.

The HPV subtypes which cause cutaneous verrucae are spread by contact between skin with microscopic or macroscopic epidermal damage and a fomite-harboring HPV. The prototypical location for contracting warts of the feet is a locker room.

Both low-risk and high-risk HPV (sometimes referred to as alpha-papillomaviruses) are considered sexually transmitted but may be spread by other forms of intimate contact. According to the Center for Disease Control and Prevention (CDC), the most recent studies show the prevalence of genital HPV for adults aged 18 to 59 to be approximately 45.2% in men and 39.9% in women

1.2.3. The landscape of HPV vaccination practice

HPV-associated neoplasia accounts for nearly one-third of all infection-attributable cancer cases. Although prevention efforts in the 20th century were largely based on cytology-based screening, current and future HPV-related cancer prevention is primarily focused on HPV vaccination and molecular screening tests. Consequently, prevention of persistent HR-HPV infections reduces the incidence of cervical cancer, and vaccination is the most economical and effective method of prevention. According to recent reports, 148 WHO member states have incorporated the HPV vaccine into their national immunization programs. 132 countries reported HPV vaccination coverage estimates for the first dose and 129 countries reported full doses in 2023, and the weighted average coverage of the first and full dose of HPV vaccination in girls aged 9-14 years were 61.6% and 47.6% respectively. Among 132 countries, fifteen countries (11%) achieved 90% coverage for the first dose. Upper-middle income countries presented the highest coverage, with 71.7% for the first dose. Global HPV vaccination coverage remains significantly below the target of 90%. While global coverage has shown a resilient recovery post-pandemic, a significant disparity remains between High-Income Countries (HICs), which benefit from robust school-based programs and digital reminder systems, and Low- and Middle-Income Countries (LMICs), where the burden of cervical cancer is highest but vaccine access remains constrained by supply chain logistics and high costs.

In Vietnam, the planned integration of HPV vaccination into the Expanded Program on Immunization from 2026 represents an important opportunity to improve equitable access. However, the current roadmap primarily targets girls aged 11 years. Therefore, while this policy may benefit future adolescent cohorts, it may not directly address the unmet vaccination needs of women aged 15–29 years examined in this study. Additional catch-up strategies, subsidized vaccination, or targeted outreach may be needed for older adolescents and young adult women who have already passed the routine vaccination age.

The evolution of the global vaccine portfolio has been instrumental in addressing these barriers. While the nonavalent vaccine (9vHPV) remains the 'gold standard' for its broad protection, the emergence of cost-effective alternatives from China (Cecolin) and India (Cervavac) has diversified the market. These newer generations, often utilizing more affordable manufacturing platforms, have significantly increased global supply. Furthermore, the paradigm-shifting recommendation for a **single-dose schedule** (off-label or official, depending on national guidelines) has gained definitive momentum. By early 2026, reinforced evidence confirmed that a single dose provides comparable efficacy to multi-dose regimens for adolescents aged 9–14.

1.3 HPV Preventive Measures

1.3.1 Strategies for HPV Prevention

Driven by the World Health Organization’s (WHO) '90-70-90' targets, global HPV prevention and vaccination strategies are undergoing a transformative shift to ensure these 2030 goals are met. Despite identifying the etiological agent, witnessing the success of screening programs in developed nations, and having effective vaccines available, the global fight against this preventable disease has yielded suboptimal results.

In November 2020, the WHO launched a global strategy to eliminate cervical cancer, aiming for an incidence rate of less than 4 per 100,000 women-years. This ambitious plan rests on three pillars: vaccinating 90% of girls by age 15, screening 70% of women at ages 35 and 45 using high-sensitivity HPV tests, and ensuring 90% of those diagnosed with cervical dysplasia or invasive cancer receive professional treatment..

Beside that, the landscape of HPV prevention is being radically transformed by cutting-edge technology and inclusive behavioral strategies. On the technological front, innovative delivery systems such as microneedle patches (MAPs) are emerging as a game-changer; these painless, needle-free alternatives offer a solution for needle phobia and, more importantly, are often thermostable. This eliminates the strict requirement for cold-chain logistics, making vaccination feasible in the most remote, resource-limited areas. Furthermore, the integration of Artificial Intelligence (AI) and Big Data allows health authorities to transition from generic to precision public health. By analyzing vast datasets, AI can predict geographical pockets with low coverage and deploy personalized digital reminders, significantly improving series completion rates.

Moreover, modern prevention strategies have shifted toward comprehensive social and behavioral interventions. Addressing vaccine hesitancy remains a top priority, with targeted campaigns designed to debunk persistent myths regarding infertility and long-term side effects. Moreover, there is a vital movement toward gender-neutral and inclusive education. By expanding the narrative beyond cervical cancer to include risks such as oropharyngeal and anal cancers, health programs are effectively engaging the LGBTQ+ community and men. This review aims to provide an updated assessment of primary and secondary prevention strategies. While time-trend analyses indicate declining incidence rates in many countries, identifying and implementing effective interventions to enhance vaccine coverage remains critical to closing the remaining public health gaps

1.3.2 Barriers and facilitators of HPV vaccination

Three prophylactic HPV vaccines have been licensed and are currently available worldwide:

• Bivalent vaccine (Cervarix): targets HPV types 16 and 18.

• Quadrivalent vaccine (Gardasil): targets HPV types 6, 11, 16, and 18.

• Nonavalent vaccine (Gardasil 9): targets HPV types 6, 11, 16, 18, 31, 33, 45, 52, and 58, potentially preventing up to 90% of cervical cancers.

The barriers reported were centralized around lack of knowledge about HPV and the HPV vaccine, fear about the safety and efficacy of the HPV vaccine, fear about not being able to pay for the HPV vaccine, and discrimination regarding the HPV vaccine. The facilitators reported were centralized around trust in the efficacy and safety of the HPV vaccine, discounted price of vaccination, positive recommendations from others, perceived risk of HPV infection, and benefits of vaccines. In other researches in India, key barriers identified were cost, lack of awareness, misinformation, and sociocultural factors. Facilitators included educational interventions, physician recommendations, and subsidized vaccination programs. School-based campaigns and community engagement were effective in improving vaccine acceptance. However, marginalized populations and out-of-school girls remain underserved. Addressing economic and sociocultural barriers, enhancing awareness, and expanding targeted vaccination programs are essential to improve HPV vaccination rates.

In Vietnam, the high cost of these vaccines-ranging from approximately 1.7 million Vietnam dong to 3.3 million Vietnam dong per dose-posed a substantial financial barrier for families in rural and underserved regions.

1.3.3 Evidence on determinants of HPV awareness and vaccination

A growing body of literature has examined factors associated with HPV vaccine awareness and uptake in LMIC settings. Key findings from systematic reviews and country-specific studies include:

Socioeconomic status. Higher household wealth is consistently associated with greater awareness and higher self-reported vaccination uptake. Studies in sub-Saharan Africa, South-East Asia, and Latin America have documented wealth-related gradients of 2–5 fold in vaccination uptake [24, 28].

Education. Women with secondary or higher education are significantly more likely to have heard of HPV vaccination and to have been vaccinated, reflecting both greater health literacy and enhanced access to information [6, 29].

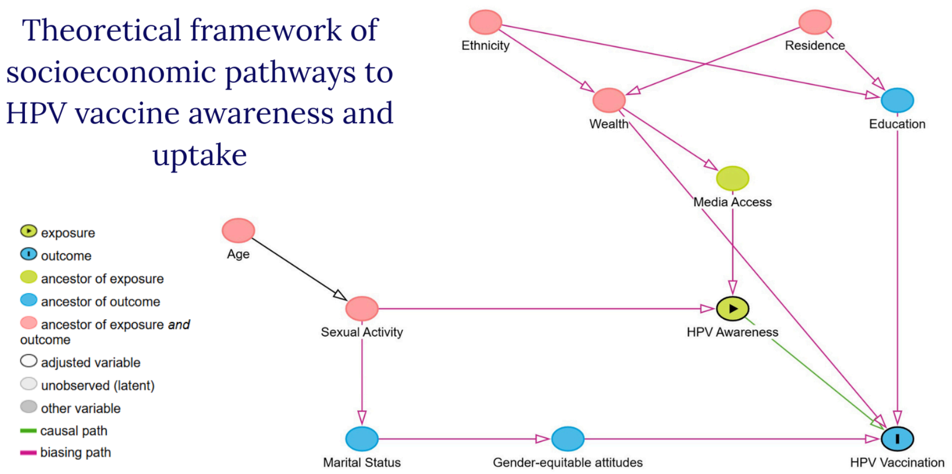
Urban–rural residence. Urban women generally have higher awareness and vaccination rates, likely due to proximity to health facilities, greater media exposure, and higher density of vaccination service points [7, 20].

Ethnicity. In multi-ethnic societies, ethnic minority women often exhibit lower awareness and uptake, attributable to language barriers, cultural beliefs, geographic isolation, and systemic marginalization [8].

Mass media and ICT exposure. Exposure to newspapers, radio, television, internet, and mobile communication has been positively associated with health knowl- edge and preventive behavior uptake in numerous health domains, including vaccination [30, 31].

Age and marital status. Younger women and those who are unmarried may be more receptive to vaccination messages, though actual uptake patterns vary by country and program design

1.3.4 DAG diagram



1.

Directed acyclic graphs (DAGs) provide a conceptual tool for organizing assumed relationships among socioeconomic position, information access, healthcare access, and HPV vaccination behavior. In this study, the DAG was used to guide covariate selection and interpretation of associations between sociodemographic factors and two main outcomes: HPV vaccine awareness and self-reported HPV vaccination uptake.

The proposed framework assumes that structural socioeconomic factors, including household wealth, educational attainment, ethnicity, and place of residence, may influence HPV vaccine awareness and uptake through several intermediary pathways. Education may affect awareness by increasing health literacy and access to reliable vaccine-related information. Household wealth may influence uptake by affecting affordability of vaccination, ability to access private vaccination services, and broader healthcare-seeking opportunities. Residence and ethnicity may shape access to health information and services through geographic availability, language, cultural acceptability, and communication barriers.

Mass media and ICT exposure, health insurance status, gender-equitable attitudes, marital status, age, and sexual experience were considered as individual or intermediary factors that may be associated with awareness, perceived need, or vaccination behavior. HPV vaccine awareness may also act as an intermediate step between socioeconomic position and vaccination uptake, because women cannot seek vaccination if they have never heard of the vaccine. However, awareness alone may not be sufficient for uptake when financial or service-access barriers remain.

Because the study used cross-sectional survey data, the DAG should be interpreted as a theoretical framework rather than proof of causal mechanisms. It supports the selection of relevant covariates for multivariable regression and provides a basis for interpreting socioeconomic inequalities in HPV vaccine awareness, uptake, and the awareness–vaccination gap.

1.4. Socioeconomic Inequalities in Healthcare

1.4.1. Concepts of Health Equity and Inequality.

In the field of population health, distinguishing between health inequality, health inequity, and health equity is essential for developing effective interventions and policies. Health inequality refers to measurable differences in health status, health outcomes, or access to healthcare between population groups. These differences may result from biological, social, economic, geographic, or health-system factors. However, not all inequalities are considered unjust; some may arise from unavoidable biological or genetic differences. Health inequity, in contrast, refers to differences that are avoidable, unfair, and unjust. In low- and middle-income countries (LMICs), socioeconomic and geographic inequalities in access to preventive and treatment services often represent important forms of health inequity, as they reflect unequal opportunities to obtain timely, affordable, and effective healthcare.

Health equity is therefore an ethical concept grounded in social justice and fairness. It refers to the absence of avoidable or remediable differences in health among groups defined by social, economic, demographic, or geographic characteristics. For many LMICs, the major challenge is not only to measure differences in health outcomes but also to identify and address the unfair conditions that produce them. Health inequities in these settings are often reflected in the unequal distribution of healthcare resources, treatment options, trained health professionals, and research opportunities. People from disadvantaged socioeconomic backgrounds may experience longer waiting times, limited access to qualified providers, and inadequate funding for diseases that disproportionately affect their communities. Addressing these gaps requires a rights-based approach to healthcare, with priority given to the fair allocation of limited resources in order to reduce avoidable and unjust disparities in health access and outcomes.g

1.4.2 The concentration index

The concentration index (CI) is the most widely used summary measure of socioeconomic inequality in health. Formally, the CI is defined as twice the area between the concentration curve and the 45-degree line of equality

$$CI = 2/\mu \times \text{cov}(y_i, r_i) \quad (1.1)$$

where y_i is the health variable for individual i, r_i is the fractional rank of individual i in the socioeconomic distribution, and μ is the population mean of y. The CI ranges from −1 to +1. A positive value indicates that the health variable is concentrated among the wealthier (pro-rich inequality), a negative value indicates concentration among the poorer (pro-poor inequality), and zero implies perfect equality

1.4.3 The Erreygers correction and decomposition

For bounded health variables (such as binary indicators of vaccination status), the standard CI may yield misleading comparisons across populations with different means. Erreygers (2009) proposed a corrected concentration index that satisfies mirror, level independence, and cardinal invariance properties

$$E(y) = 4\mu\, b - a \cdot CI \quad (1.2)$$

where a and b are the lower and upper bounds of the health variable (0 and 1 for binary outcomes). [8] The Erreygers-corrected CI can be decomposed into the contributions of individual determinants using the relationship:

$$E(y) = 4X\ k\ \beta_k \bar{x}_k \cdot CI_k + \text{residual} \quad (1.3)$$

where β_k is the marginal effect of covariate x_k from a probit model, $\bar{x}_k$ is the covariate mean, and CI_k is the concentration index of the covariate with respect to socioeconomic rank. This decomposition enables researchers to quantify the percentage contribution of each factor—such as education, wealth, or geographic location—to the total observed inequality

CHAPTER II: MATERIALS AND METHODS

2.1. Study design

The data for this study were drawn from the Survey measuring Viet Nam Sustainable Development Goal indicators on Children and Women (SDGCW/MICS) 2020-2021. The survey employed a two-stage, stratified cluster sampling design utilizing a sampling frame based on the 2019 Viet Nam Census of Population and Housing. The sample was stratified by urban and rural areas, six socio-economic regions, and two major cities (Ha Noi and Ho Chi Minh City) as the primary sampling domains. Within each domain, census enumeration areas were further stratified by high and low proportions of ethnic minorities. In the first stage, 700 enumeration areas (clusters) were systematically selected with probability proportional to size. In the second stage, a systematic sample of 20 households was drawn from each selected enumeration area, yielding a total sample of 14,000 households. For the women's module, the Individual Questionnaire for Women was administered to all eligible women aged 15 to 49 years residing in all the selected households. Out of 11,294 eligible women identified in the successfully interviewed households, 10,770 were successfully interviewed, resulting in a specific response rate of 95.4% and an overall response rate of 94.3%. Finally, because the sample is not self-weighting, sample weights were applied in all statistical analyses to ensure accurate estimation and national representativeness.

2.2. Study participants

2.2.1 Inclusion criteria

• Women aged 15–29 years: who completed the Women's Individual Questionnaire.

• Valid sampling weights, primary sampling unit (PSU), and stratification information available for survey-weighted analysis.

2.2.2 Exclusion criteria

• Women with missing sampling weights (wmweight), PSU identifiers, or stratum indicators, as these preclude valid survey-weighted estimation.

• Records with indeterminate responses on both outcome variables (HPV awareness and vaccination status)

2.3 Sample size

The analytic sample consisted of all eligible women from the MICS6 survey who met the inclusion criteria. After applying exclusion criteria, the final sample comprised N = 4,102 women aged 15–29 years. This sample size provides adequate statistical power for detecting inequality patterns and conducting multivariable regression and decomposition analyses. No additional sample size calculation was performed, as this study utilized the complete available dataset from a nationally representative survey.

2.4 Study variables

2.4.1 Outcome variables

Three binary outcome variables were constructed from the MICS6 cervical cancer prevention module:

1. HPV vaccine awareness (ccp5_bin): Derived from MICS item CCP5 ("Have you ever heard of HPV vaccination?"). Coded as 1 = Yes, 0 = No.

2. HPV vaccination status (ccp6_bin): Derived from MICS item CCP6 ("Have you ever received the HPV vaccine?"). Coded as 1 = Yes, 0 = No or Don't Know. Women who reported not having heard of HPV vaccination (CCP5 = No) were classified as unvaccinated (CCP6 = 0), consistent with the MICS skip pattern.

3. Awareness–vaccination gap (gap): Defined among aware women only (CCP5 = Yes). Coded as 1 if the woman was aware but unvaccinated, 0 if aware and vaccinated.

2.4.2 Explanatory variables

This variable captures the "knowing–doing gap" relevant to behavioral intervention design based on the conceptual framework (Section 1.3.3) and available MICS6 data, the following covariates.

Wealth index quintiles. The MICS wealth index is an asset-based compos-ite measure of household socioeconomic status. It is constructed from information on household assets (e.g., ownership of durable goods such as television, refrigera- tor, motorcycle), dwelling characteristics (e.g., flooring and roofing materials), and access to basic services (e.g., water source, sanitation facility type). These binary indicators are synthesized using **principal component analysis (PCA)** to generate a continuous wealth score for each household. Households are then ranked by this score and divided into five equal-sized groups (quintiles), from Q1 (poorest 20%) to Q5 (richest 20%). The categorical wealth quintile variable was used in descriptive, bivariate, regression, and decomposition analyses; the continuous wealth score was used for socioeconomic ranking in concentration index estimation.

2.5. Data analysis

All analyses were performed using Stata version 19SE (StataCorp LLC, College Station, TX) with the svy: prefix to account for the complex survey design (stratification, clustering, and sampling weights). The analytical strategy followed a sequential six-step approach

Step 1: Descriptive statistics Survey-weighted proportions with 95% confidence intervals were computed for all outcome and explanatory variables to characterize the study population and estimate the prevalence of HPV awareness and vaccination.

Step 2: Bivariate analysis The association between each explanatory variable and the outcome variables was assessed using survey-weighted Pearson chi-squared tests with RaoScott corrections for complex survey design. Row percentages with 95% confidence intervals were reported. Statistical significance was set at $\alpha = 0.05$.

Step 3: Concentration index The Wagstaff concentration index (CI) provides a rigorous econometric quantification of socioeconomic inequality in health outcomes [23]. In this study, the standard CI was formulated mathematically as both the scaled covariance between the health outcome and the fractional

rank in the living standards distribution, and geometrically as the area integral:
$CI = 2/\mu \times cov(y_i, r_i)$ (2.1)
where yi represents the binary outcome for individual i, ri is the continuous fractional rank in the socioeconomic distribution (household wealth score), and μ is the surveyweighted population mean of the outcome. Geometrically, the CI equals twice the area between the concentration curve and the line of equality. A positive CI ($0 < 17$ $CI \leq 1$) indicates pro-rich inequality, whereas a negative CI ($-1 \leq CI < 0$) signifies pro-poor concentration.
Step 4: Multivariable Poisson regression Because the prevalence of HPV awareness exceeded the 10% threshold, logistic regression would mathematically yield odds ratios that severely overestimate the true relative risk. Consequently, we employed modified Poisson regression—a generalized linear model (GLM) featuring a logarithmic link function and a Poisson distribution—to directly estimate adjusted prevalence ratios (PR). To correct for the violation of the Poisson assumption (variance equal to the mean) inherent in binary data, robust Huber-White sandwich variance estimators were applied.
The formal model specification is: $\log E(Y_i \mid X_i) = X^\top_i \beta$ with robust variance: $Vb(\hat{\beta}) = D^{-1}MD^{-1}$ (2.2) where $E(Y_i \mid X_i)$ is the expected probability of the outcome given the covariate vector Xi , and β is the vector of coefficients. The exponentiated coefficients ($\exp(\beta_k)$) provide the direct prevalence ratios. The matrix D represents the expected information matrix, and M is the outer product of the empirical gradients. The reference categories were: 15–19 age, primary or less, Q1 poorest , currently married and the absence category for binary covariates. A sensitivity analysis using logistic regression (yielding odds ratios) was conducted to assess the

robustness of findings. For HPV vaccination uptake, since the prevalence was below 10%, multivariable logistic regression was applied to estimate adjusted Odds Ratios (aOR).

Step 5: Erreygers decomposition A fundamental mathematical limitation of the standard Wagstaff CI is that its absolute bounds contract as the mean of the binary outcome deviates from 0.5, distorting comparative analyses across populations. To circumvent this scale dependence, we utilized the Erreygers Concentration Index (ECI), which restores the normative 18 limits to $[-1, 1]$ via the transformation:

$$ECI(y)=4\mu(1-\mu)CI(y) \quad (2.3)$$

The ECI was subsequently decomposed to isolate the specific econometric contributions of each explanatory variable [7]. Given a generalized linear mapping via a probit link, the exact contribution of each covariate k is calculated as the product of its partial marginal effect, its weighted mean, and its own concentration index:

$$Contribution_k = 4 \cdot \frac{\partial E(y \mid x)}{\partial x_k} \cdot \bar{x}_k \cdot CI_{x_k} = 4 \cdot \beta_k^{AME} \cdot \bar{x}_k \cdot CI_{x_k} \quad (2.4)$$

where β_k^{AME} is the average marginal effect from a probit model, $\bar{x}_k$ is the weighted mean of the covariate, and CI_{x_k} is the Erreygers-corrected concentration index of the covariate with respect to the wealth ranking. The percentage contribution of each covariate was computed as:

$$Percentage_k = \frac{Contribution_k}{ECI_{total}} \times 100 \quad (2.5)$$

Decomposition was performed separately for each of the three outcome variables: awareness, vaccination, and the awareness–vaccination gap.

Step 6: Stratified and sensitivity analyses Additional analyses included:

• Stratified regression by urban/rural residence to examine whether determinants differed between settings.

• Crude (unadjusted) Poisson regression for each variable individually to compare crude and adjusted prevalence ratios.

• Logistic regression sensitivity analysis to verify that key findings were robust 19 to the choice of link function.

• Gap model: Poisson regression among aware women to identify factors associated with failure to translate awareness into vaccination action.

2.6 Ethical considerations

The MICS6 survey protocol was approved by the Institutional Review Board of the General Statistics Office of Vietnam and received ethical clearance from UNICEF’s Division of Data, Analytics, Planning, and Monitoring. Written informed consent was obtained from all individual respondents prior to interview participation. For respondents under the age of 18, parental/guardian consent was also obtained. This study constitutes a secondary analysis of de-identified, publicly available survey data. No direct contact with human subjects occurred. All data were accessed and analyzed in compliance with the MICS data use agreement.

CHAPTER III: RESULTS

3.1 Characteristics of the study population

Variables	N=4102	%	% weighted
Age group			
15-19	1349	32.9	30.4
20-24	1150	28.0	29.7
25-29	1603	39.1	39.9
Residence			
Urban	1206	29.4	37.4
Rural	2896	70.6	62.6
Ethnicity			
Kinh/ Hoa	2415	58.9	85.0
Ethnic minority	1678	41.1	15.0
Marital status			
Currently married/ union	2268	55.3	46.4
Formerly married/union	106	2.6	3.3
Never married/union	1728	42.1	50.3
Education			

Primary or less	601	14.7	5.2
Lower secondary	1074	26.2	22.1
Upper secondary and tertiary	2427	59.1	72.7
Wealth quintile			
Q1 (poorest)	1587	38.7	19.2
Q2	716	17.4	21.2
Q3	647	15.8	21.9
Q4	603	14.7	20.1
Q5 (richest)	549	13.4	17.6
Health insurance			
Insured	3605	87.9	86.3
Uninsured	497	12.1	13.7
Mass media & ICT exposure			
Frequent access	3373	82.3	90.9
Infrequent access	729	17.7	9.1
Gender-equitable attitude			
Rejects all wife-beating	3367	82.1	90.2
Does not reject all	534	17.9	9.8

Sexual behavior			
Yes	2470	60.2	53.1
No	1632	39.8	46.9

Table 1. Sociodemographic and behavioral characteristics of the study population (N=4,102)

The 25–29 age group made up 39.9% of the study population, followed by 15–19 (30.4%) and 20–24 (29.7%). Rural residents (62.6%) and Kinh/Hoa ethnicities (85.0%) dominated the participants. Moreover half of respondents (50.3%) had never been married or in a relationship, while 46.4% were married or cohabiting. Only 3.3% had been married or in a relationship. The study participants were well-educated. Only 5.2% had completed primary school, while 72.7% had completed upper secondary or university.

The weighted statistics showed that wealth quintiles were evenly distributed, with 17.6% in Q5 and 21.9% in Q3. Population health insurance was 86.3%. Most (90.9%) regularly used ICT. Most (90.2%) thought gender equality was great and rejected spousal abuse arguments. According to behavioral tests, 53.1% of respondents had sex, while 46.9% had never experienced it.

3.2. Prevalence of HPV Awareness and Vaccination Practice

Outcomes variables (N=4102)			
Variables	**n**	**%**	**% weighted**
HPV vaccine awareness			
Yes	2133	52.0	62.4
No	1969	48.0	37.6

HPV vaccination			
Yes	220	5.4	7.5
No	3882	94.6	92.5

Table 2. Prevalence of HPV vaccine awareness and uptake among the study population. (N=4,102)

Regarding the primary outcomes, 62.4% of women reported having heard of HPV vaccination, while only 7.5% had received the vaccine, revealing a substantial awareness–vaccination gap.

3.3. Factors Associated with HPV Awareness and Practice

Variables	N	n	(Aware)%	95% CI	p-value
Age group					
15-19	1349	554	48.3	44.1 - 52.2	<0.001
20-24	1150	623	66.2	62.1 - 70.2	
25-29	1603	956	70.3	67.1 - 73.5	
Residence					
Urban	1206	855	73.1	69.2 - 76.9	<0.001
Rural	2896	1278	56.0	52.7 - 59.4	
Ethnicity					
Kinh/ Hoa	2415	1631	67.0	64.2 - 69.8	<0.001

Ethnic minority	1678	502	36.1	31.6 - 40.5	
Marital status					
Currently married/ union	2268	1182	63.6	60.4 - 66.7	0.270
Formerly married/union	106	52	54.1	41.0 - 66.7	
Never married/union	1728	963	61.8	58.3 - 65.2	
Education					
Primary or less	601	89	22.9	15.1–30.7	<0.001
Lower secondary	1074	431	49.2	44.2–54.1	
Upper secondary and tertiary	2427	1613	69.2	66.5–72.0	
Wealth quintile					
Q1 (poorest)	1587	414	33.3	28.8–37.8	<0.001
Q2	716	407	58.5	53.7–63.4	
Q3	647	409	64.4	59.4–69.3	
Q4	603	446	73.5	69.1–78.0	
Q5 (richest)	549	457	83.6	79.9–87.3	
Health insurance					
Insured	3605	1889	63.0	60.3–65.7	0.218
Uninsured	497	224	58.7	51.8–65.6	

Mass media & ICT exposure					
Frequent access	3373	1984	64.7	62.1–67.3	<0.001
Infrequent access	729	149	39.1	32.6–45.6	
Gender-equitable attitude					
Rejects all wife-beating	3367	1887	65.2	62.5–67.9	<0.001
Does not reject all	534	177	45.2	38.4–51.9	
Sexual behavior					
Yes	2470	1237	63.5	60.3 - 66.5	0.272
No	1632	896	61.1	57.5 - 64.6	

Table 3. Prevalence of HPV vaccine awareness by sociodemographic and behavioral characteristics. (N = 4,102)

Results indicate significant disparities ($p < 0.001$) in awareness influenced by age, residence, ethnicity, education, wealth, information accessibility, and gender-equitable attitudes. Awareness increased with age, reaching a maximum of 70.3% in the 25–29 age group and declining to 48.3% in the 15–19 age group. Urban participants exhibited more awareness (73.1%) compared to rural people (56.0%). Compared to ethnic minorities' 36.1% awareness, the Kinh/Hoa majority had 67.0%. Wealth and education increased vaccine knowledge. At 69.2%, upper secondary students were the most aware, compared to 22.9% for primary students. Knowledge rose steadily from Q1 (33.3%) to Q5 (83.6%). People with regular access to mass media and ICT and gender-equitable attitudes (refuting all rationalizations for spousal abuse) had higher awareness rates (64.7% and 65.2%, respectively) than those with limited access (39.1%) or those who did not disavow such violence (45.2%). Neither marital status (p = 0.270), health insurance coverage (p = 0.218), nor sexual conduct (p = 0.272) significantly affected HPV vaccine awareness.

Variables	**IRR**	**95% CI**	**p-value**
Age group (ref: 15-19)			
20-24	1.37	1.24 - 1.51	<0.001
25-29	1.45	1.32 - 1.59	
Residence (ref: Urban)			
Rural	0.76	0.70 - 0.83	<0.001
Ethnicity (ref: Kinh/ Hoa)			
Ethnic minority	0.54	0.47 - 0.61	<0.001
Marital status (ref: Currently married/ union)			
Formerly married/union	0.85	0.67 - 1.08	0.185
Never married/union	0.97	0.90 - 1.04	0.419
Education (ref: Primary or less)			
Lower secondary	2.15	1.51 - 3.03	<0.001
Upper secondary and tertiary	3.02	2.14 - 4.27	
Wealth quintile (ref: Q1 (poorest))			

Q2	1.75	1.51- 2.10	<0.001
Q3	1.93	1.66 - 2.25	
Q4	2.21	1.91 - 2.55	
Q5 (richest)	2.51	2.18 - 2.89	
Health insurance (ref: Insured)			
Uninsured	0.93	0.82 - 1.05	0.254
Mass media & ICT exposure (ref: Infrequent access)			
Frequent access	1.66	1.41 - 1.97	<0.001
Gender-equitable attitude (ref: Does not reject all)			
Rejects all wife-beating	1.44	1.26 - 1.64	<0.001
Sexual behavior (ref: No)			
Yes	1.03	0.97 - 1.11	0.274

Table 4. Unadjusted associations between sociodemographic characteristics and HPV awareness (N=4,102)

The studies identify several characteristics that are important in awareness of the HPV vaccine. All positively linked characteristics were statistically significant ($p < 0.001$). Awareness is positively associated with age; the age groups 20-24 and 25-29 are 1.37 and 1.45 times more aware than the age group 15-19 accordingly. Educational level showed a gradient effect, with persons with lower secondary education 2.15 times more likely to be conscious than those with elementary education or less. Upper secondary education or higher was associated with a 3.02 likelihood. There was a significant association between economic status and awareness that increased throughout wealth quintiles from Q1 to Q5 with the wealthiest having the highest level of knowledge (IRR = 2.51; 95% CI: 2.18–2.89) . Media and ICT users were 1.66 times more aware while gender equality campaigners like those against domestic abuse were 1.44 times more aware. However, HPV vaccine knowledge was significantly influenced by a variety of circumstances. Knowledge of rural inhabitants was considerably lower, being 0.76 times lower than urban inhabitants (IRR = 0.76; 95% CI: 0.70–0.83; $p < 0.001$). Awareness levels were lower in ethnic minority groups compared to the mainstream Kinh/Hoa population ($p < 0.001$).

Finally, HPV vaccine awareness was not significantly correlated with marital status, health insurance coverage, and sexual conduct. No significant differences were found for formerly married ($p = 0.185$) or never married ($p = 0.419$) individuals compared to current marriage, and no effect was noted for health insurance (IRR = 0.93; $p = 0.254$) and sexual experience (IRR = 1.03; $p = 0.274$) on awareness.

Variables	**N**	**n**	**(Practice)%**	**95% CI**	**p-value**
Age group					
15-19	1349	57	5.6	3.6 - 7.7	0.015
20-24	1150	76	10.0	7.5 - 12.5	

25-29	1603	87	7.0	5.3 - 8.7	
Residence					
Urban	1206	125	11.4	8.9 - 13.9	<0.001
Rural	2896	95	5.1	3.8 - 6.5	
Ethnicity					
Kinh/ Hoa	2415	203	8.5	7.0 -10.0	<0.001
Ethnic minority	1678	17	1.6	0.4 - 2.8	
Marital status					
Currently married/ union	2268	85	5.1	3.9 - 6.5	
Formerly married/union	106	8	7.3	3.4 - 15.0	<0.001
Never married/union	1728	127	9.7	7.7 - 12.1	
Education					
Primary or less	601	3	0.8	0.1 - 4.0	
Lower secondary	1074	17	2.4	1.3 - 4.3	<0.001
Upper secondary and tertiary	2427	200	9.5	8.0 - 11.3	
Wealth quintile					
Q1 (poorest)	1587	11	0.7	0.1 - 1.3	<0.001
Q2	716	31	6.0	3.7 - 8.4	

Q3	647	34	5.9	3.8 - 8.1	
Q4	603	56	9.2	6.3 - 12.0	
Q5 (richest)	549	88	16.6	12.3 - 20.8	
Health insurance					
Insured	3605	204	8.0	6.7 - 9.5	0.056
Uninsured	497	16	4.0	2.3 - 7.0	
Mass media & ICT exposure					
Frequent access	3373	213	8.3	6.7 - 9.4	0.017
Infrequent access	729	7	2.6	1.0 - 6.9	
Gender-equitable attitude					
Rejects all wife-beating	3367	198	7.9	6.7 - 9.4	0.050
Does not reject all	534	22	4.7	2.8 - 7.9	
Sexual behavior					
Yes	2470	101	5.4	4.3 - 6.8	<0.001
No	1632	119	9.8	7.8 - 12.3	

Table 5. Prevalence of HPV vaccine practice by sociodemographic and behavioral characteristics (N=4102)

Demographic, socioeconomic, and behavioral factors strongly influence HPV vaccination habit, according to the study. The greatest vaccination rate was

10.0% in the 20–24 age group (p=0.015), followed by 7.0% in the 25–29 group and 5.6% in the 15–19 group. Vaccination rates differed by place of residence and ethnicity ($p < 0.001$), with urban residents (11.4%) having twice the rate of rural residents (5.1%) and the Kinh/Hoa group (8.5%) having a significantly higher rate than ethnic minorities (1.6%). A significant predictor was marital status ($p < 0.001$), with the greatest immunization rate (9.7%) in the never-married group, followed by formerly married (7.3%) and currently married or cohabiting (5.1%). Additionally, substantial differences were seen in education and economic position ($p<0.001$). Upper secondary and higher education students had a 9.5% immunization rate, while elementary school students had 0.8%. Vaccination rates increased from 0.7% in Q1 to 16.6% in Q5. Higher HPV vaccination rates were linked to regular media and ICT exposure (8.3% vs. 2.6%; $p = 0.017$), gender equality attitudes (7.9% vs. 4.7%; $p = 0.050$), and no history of sexual activity (9.8% vs. 5.4%; $p < 0.001$). The study found non-statistically significant immunization practice variables. Health insurance doubled the vaccination rate of uninsured people (4.0%), however the difference was not statistically significant ($p = 0.056$).

Variables	**IRR**	**95% CI**	**p-value**
Age group (ref: 15-19)			
20-24	1.77	1.19 - 2.63	0.005
25-29	1.24	0.81 - 1.92	0.317
Residence (ref: Urban)			

Rural	0.45	0.31 - 0.64	<0.001
Ethnicity (ref: Kinh/ Hoa)			
Ethnic minority	0.19	0.08 - 0.41	<0.001
Marital status (ref: Currently married/ union)			
Formerly married/union	1.43	0.64 - 3.18	0.375
Never married/union	1.92	1.37 - 2.68	<0.001
Education (ref: Primary or less)			
Lower secondary	3.09	0.53 - 18.05	0.209
Upper secondary and tertiary	12.19	2.28 - 65.03	0.03
Wealth quintile (ref: Q1 (poorest))			
Q2	8.58	3.28 - 22.50	<0.001
Q3	8.43	3.30 - 21.54	
Q4	13.01	5.22 - 32.38	
Q5 (richest)	23.51	9.58 - 57.62	
Health insurance (ref: Insured)			
Uninsured	0.50	0.28 - 0.89	0.020
Mass media & ICT exposure (ref: Infrequent access)			
Frequent access	3.03	1.13 - 8.13	0.027

Gender-equitable attitude (ref: Does not reject all)			
Rejects all wife-beating	1.67	0.98 - 2.82	0.056
Sexual behavior (ref: No)			
Yes	0.55	0.41 - 0.76	<0.001

Table 6. Unadjusted associations between sociodemographic characteristics and HPV status (N=4102)

The unadjusted analysis reveals a high connection between HPV vaccination status and other socio-demographic, socio-economic and behavioral factors. The 20-24 age group was 1.77 times more likely to be vaccinated than the 15-19 age group ($p = 0.005$). Compared with their urban and Kinh/Hoa counterparts, individuals living in rural areas and ethnic minorities had significantly lower likelihood of immunization (IRR = 0.45 and 0.19, respectively, $p < 0.001$). The never-married cohort had some 2 times the odds of being vaccinated (IRR = 1.92; $p < 0.001$) compared to those who are now married. The level of education and income considerably improved the vaccination rate; individuals with upper secondary education or above were 12.19 times more likely to get vaccinated than those with elementary education or less ($p = 0.03$). The probability of vaccination increased significantly across wealth quintiles. The wealthiest quintile (Q5) was 23.51 times more likely to be vaccinated than the poorest (Q1) ($p < 0.001$). Higher likelihood of vaccination was associated with regular exposure to mass media and ICT (IRR = 3.03; $p = 0.027$) whereas lower likelihood of vaccination was associated with absence of insurance (IRR = 0.50; $p = 0.020$) and previous sexual activity (IRR = 0.55; $p < 0.001$). No statistically significant associations were seen between vaccination status and the 25-29 age cohort (p = 0.317), previously married status (p = 0.375), lower secondary education (p = 0.209), or attitudes that completely reject domestic violence (p = 0.056).

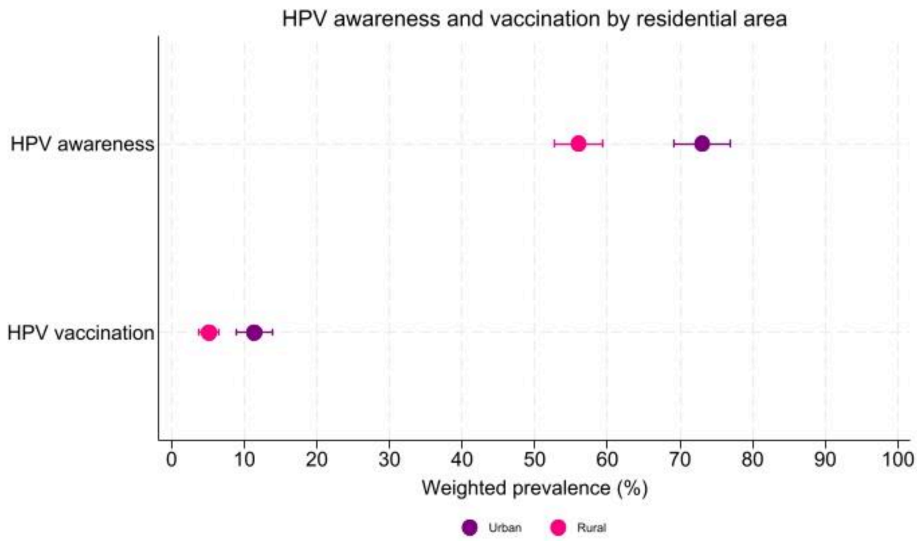


Figure 1. Prevalence of HPV vaccine awareness and vaccination practice by residential area.

Figure 1 shows a big difference between urban and rural areas. Compared to rural participants (awareness below 60% and vaccination around 5%), urban participants had awareness above 70% and vaccination rates around 10%. Thus, the non-overlapping error bars prove this difference is significant.

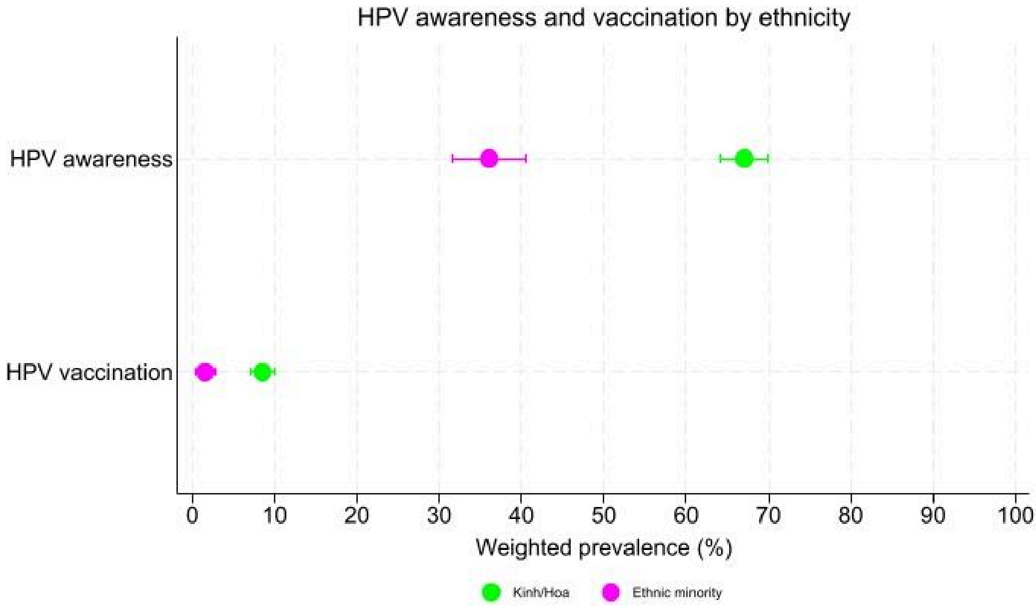


Figure 2. Prevalence of HPV vaccine awareness and vaccination practice by ethnicity.

Figure 2 illustrates pronounced disparities in health information access and healthcare utilization across ethnic groups. Specifically, the awareness rate among the Kinh/Hoa ethnic majority approached 70%, whereas it stood at just over 30% for ethnic minority populations. Furthermore, while Kinh/Hoa vaccination rates approached 10%, ethnic minority vaccination rates were approximately 0%.

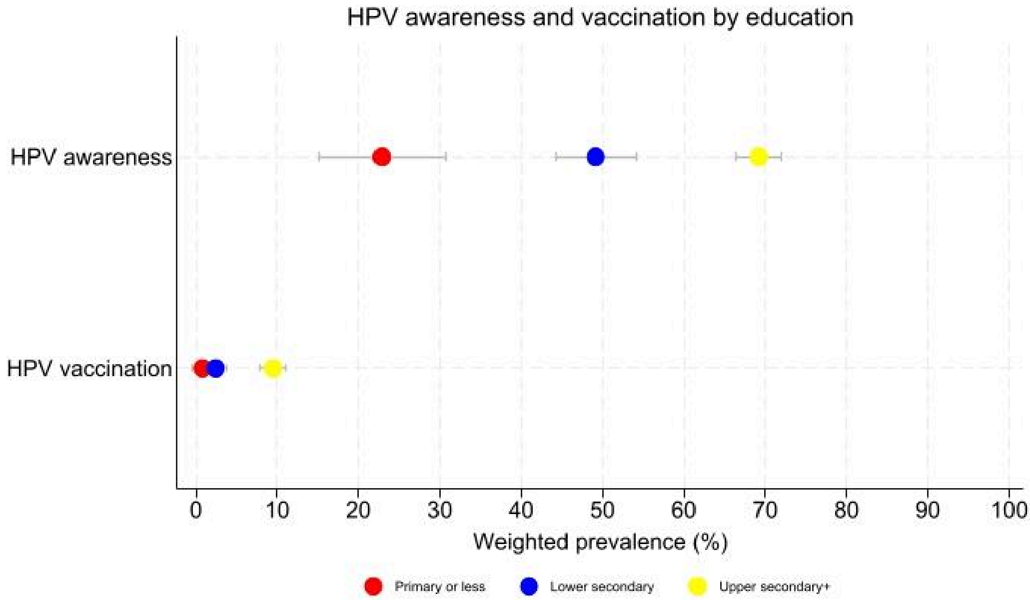


Figure 3. Prevalence of HPV vaccine awareness and uptake by educational attainment.

Figure 3 demonstrates a clear positive gradient between educational attainment and HPV vaccine-related outcomes. Specifically, advanced education is associated with elevated levels of both awareness and vaccination uptake, peaking among individuals with an upper secondary education or higher. Conversely, the cohort with primary education or less exhibited the lowest prevalences for both metrics. Notably, even within the highest educational tier, actual vaccination rates remained below the 10% threshold.

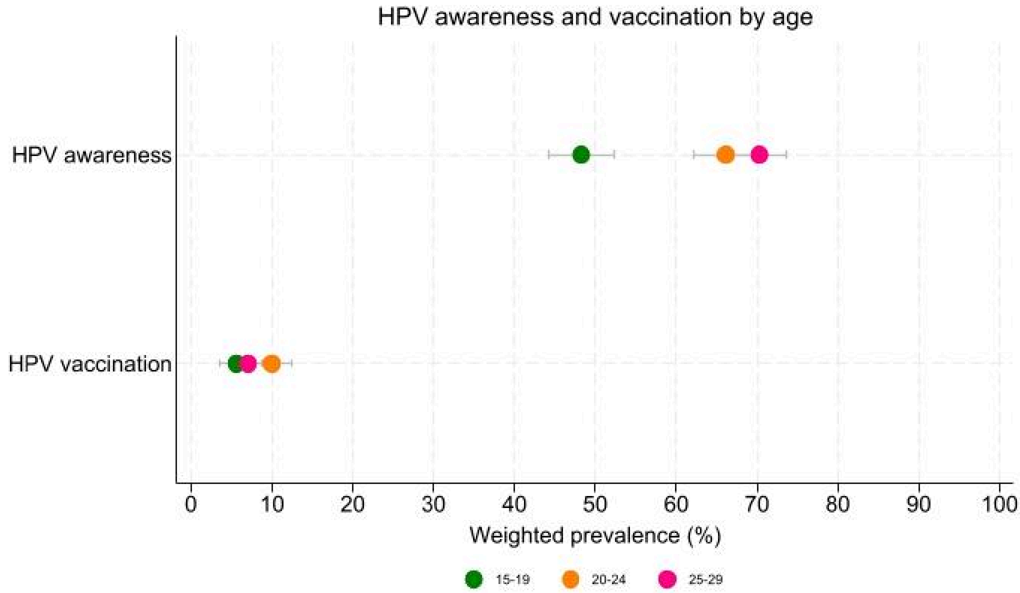


Figure 4. Age-group HPV vaccination awareness and uptake.

Figure 4 shows a significant gap between health awareness and immunization across age cohorts. Awareness prevalence rises with age, highest in 25–29, followed by 20–24, and lowest in 15–19. In terms of actual vaccination uptake, the 20–24 age group leads (around 10%), exceeding the 25–29 age group. In all cognitive and behavioral metrics, adolescents (15–19 years) had the lowest prevalences.

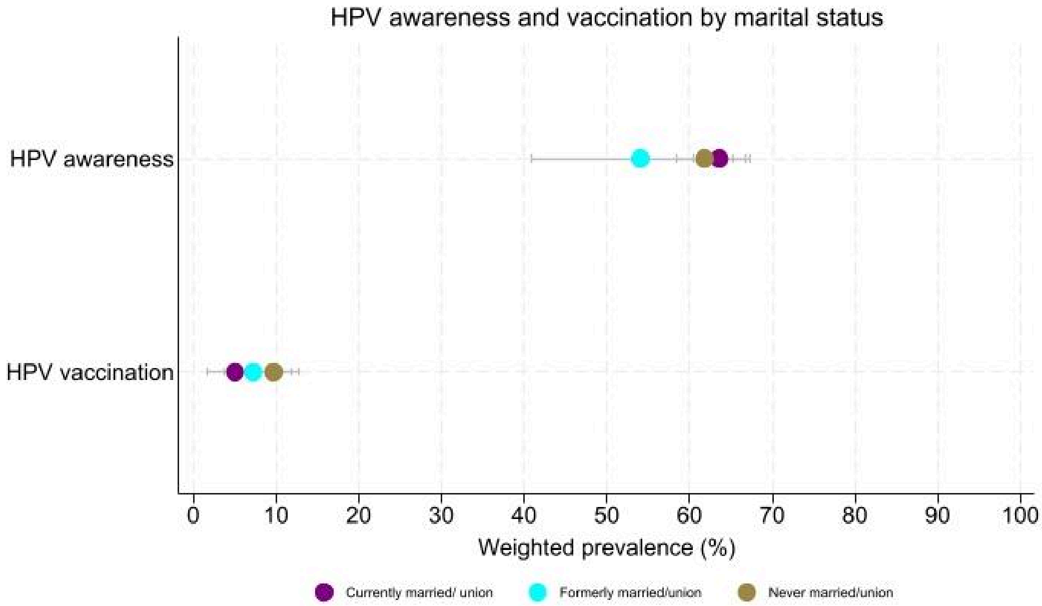


Figure 5. HPV vaccination awareness and uptake by marital status.

Figure 5 indicates that married people and single people are equally aware. Vaccination uptake is highest in the never-married/union group (nearly 10%) and lowest in the already married group (5%). As predicted, the never-married cohort is mostly younger people in the main target age range, indicating a higher need for HPV vaccination.

3.4. Socioeconomic Inequalities in HPV Prevention

3.4.1 Overall concentration indices

Outcome	Index	Estimate	SE	95%CI	p-value
Awareness (n=4102)	Erreygers	0.371	0.022	0.328-0.415	<0.001
	Wagstaff	0.395	0.024	0.349-0.442	<0.001
Vaccination	Erreygers	0.116	0.014	0.089-0.144	<0.001

(n=4102)	Wagstaff	0.421	0.051	0.320-0.521	<0.001
Gap (n=2133)	Erreygers	-0.123	0.021	-0.164 - -0.082	<0.001
	Wagstaff	-0.292	0.050	-0.389 - -0.195	<0.001

Table 7. Presents the overall concentration indices for each outcome by Erreygers and Wagstaff index

The concentration indices indicated different directions of socioeconomic inequality across the three outcomes. HPV vaccine awareness and vaccination uptake were positively concentrated among wealthier women, indicating pro-rich inequality. In contrast, the awareness–vaccination gap showed a negative concentration index (Erreygers CI = −0.123; Wagstaff CI = −0.292; p < 0.001). Because the gap was coded as being aware but unvaccinated, this negative value indicates that, among women who had heard of HPV vaccination, failure to translate awareness into vaccination was disproportionately concentrated among poorer women.

3.4.2 Concentration curves

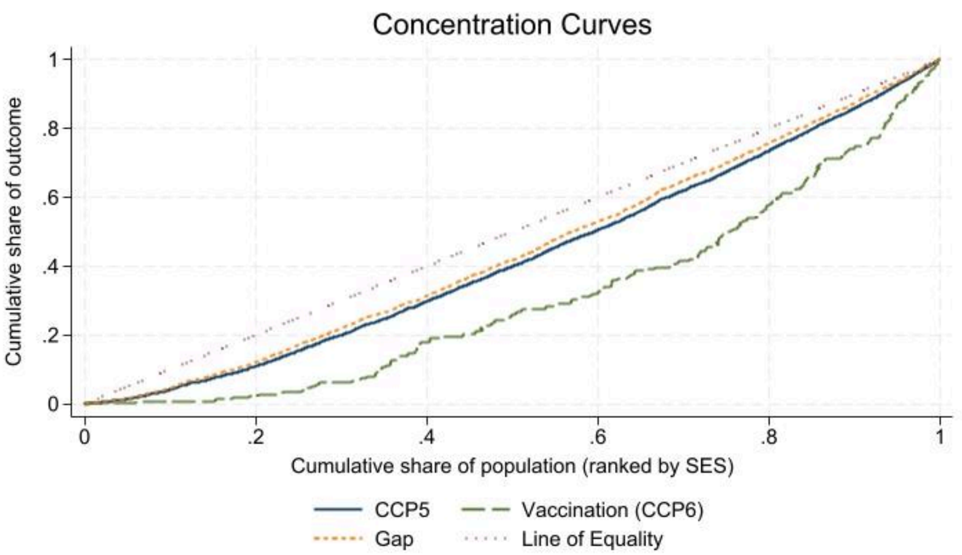


Figure 6. Concentration curves for HPV vaccine awareness, vaccination uptake, and the knowledge-practice gap.

Concentration curves were used to look at the socioeconomic distribution of HPV-related outcomes. They showed big variations in health literacy and vaccine uptake. The awareness curve (CCP5) was quite close to the line of equality, which means that information was spread out across all socioeconomic groups. However, the vaccination curve (CCP6) showed a big drop. This visual evidence confirms a significant pro-affluent bias in vaccine uptake, as the cumulative fraction of vaccinations is disproportionately concentrated among the wealthiest segments of the population. The growing gap between knowledge and action among people with low incomes shows that there is a big "knowledge-to-action" gap. This means that even if people are becoming more aware, money problems continue to be a major structural barrier to fair vaccine access in Vietnam.

Multicollinearity diagnostics revealed severe collinearity between the 'sexual behavior' variable and the 'never married/union' category, yielding Variance Inflation Factor (VIF) values exceeding 11. After excluding 'sexual behavior' from the model, all VIF values dropped to acceptable thresholds (mean VIF = 1.79), confirming the absence of problematic multicollinearity among the remaining independent variables. In the Vietnamese cultural context, premarital sex is often underreported or closely tied to marital status in cross-sectional surveys. This strong socio-cultural correlation explains the severe multicollinearity between the 'never married' and 'sexual behavior' variables, justifying the exclusion of the latter from the multivariable models. (Table 1: Variance Inflation Factor (VIF) for independent variables (Appendix))

3.4.3 Multivariable regression

Variables	Awareness			Vaccination		
	aPR	95%CI	p-value	aOR	95%CI	p-value

Age group (ref: 15-19)						
20-24	1.433	1.31 1.569	<0.001	2.547	1.619 - 4.007	<0.001
25-29	1.563	1.405 1.738	<0.001	2.192	1.238 - 3.811	0.006
Residence (ref: Urban)						
Rural	0.944	0.877 - 1.017	0.131	0.764	0.507 - 1.157	0.204
Ethnicity (ref: Kinh/ Hoa)						
Ethnic minority	0.847	0.736 - 0.974	0.020	0.683	0.268 - 1.740	0.424
Marital status (ref: Currently married/ union)						
Formerly married/union	0.876	0.725 - 1.059	0.171	1.613	0.677 - 3.845	0.280
Never married/union	1.019	0.941 - 1.102	0.648	1.842	1.165 - 2.912	0.009
Education (ref: Primary or less)						
Lower secondary	1.697	1.199 - 2.402	0.003	1.714	0.284 10.339	0.556
Upper secondary and tertiary	2.040	1.432 - 2.905	<0.001	3.366	0.600 - 18.886	0.168
Wealth quintile (ref: Q1 (poorest))						
Q2	1.402	1.19 - 1.652	<0.001	5.014	1.667 - 14.608	0.004
Q3	1.431	1.201 - 1.705	<0.001	3.962	1.314 - 11.851	0.014

Q4	1.584	1.336 - 1.878	<0.001	6.022	1.939 - 18.323	0.002
Q5 (richest)	1.776	1.489 - 2.118	<0.001	10.719	3.396 - 32.883	<0.001
Health insurance (ref: Insured)						
Uninsured	0.994	0.896 - 1.102	0.907	0.606	0.325 - 1.137	0.117
Mass media & ICT exposure (ref: Infrequent access)						
Frequent access	1.151	0.987 - 1.342	0.073	1.278	0.452 - 3.485	0.637
Gender-equitable attitude (ref: Does not reject all)						
Rejects all wife-beating	1.219	1.076 - 1.381	0.002	1.213	0.729 - 2.325	0.498

Table 8. Adjusted prevalence ratios (aPR) for awareness and adjusted odds ratios (aOR) for vaccination from multivariable regression.

For HPV vaccination uptake, survey-weighted multivariable logistic regression was used to estimate adjusted odds ratios because the outcome prevalence was below 10%. In the adjusted model, age group and household wealth remained significantly associated with HPV vaccination uptake. Women aged 20–24 and 25–29 had higher odds of reporting HPV vaccination than those aged 15–19. A strong wealth gradient persisted, with women in the richest quintile having approximately 10.7 times higher odds of vaccination than those in the poorest quintile. Never-married women also had higher adjusted odds of vaccination compared with currently married women. In contrast, residence, education, insurance status, ethnicity, mass media/ICT exposure, and gender-equitable attitudes were not statistically significant for vaccination uptake after adjustment.

3.4.4 Decomposition of awareness inequality

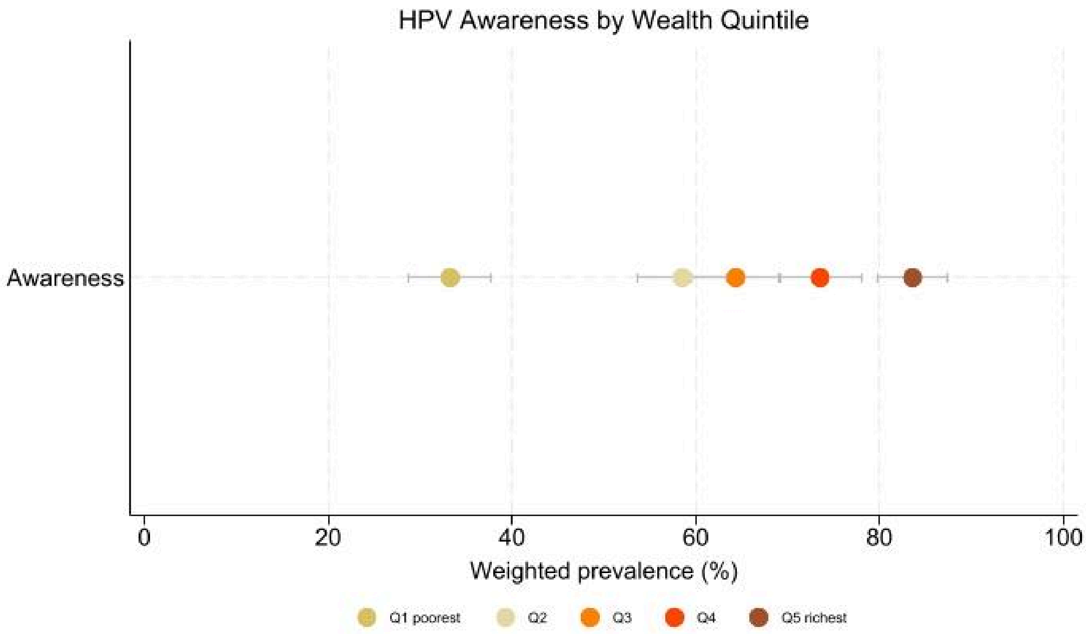


Figure 7. Awareness of the HPV vaccine and vaccination practices categorized by wealth quintile.

Figure 7 demonstrates a distinct upward trend in awareness relative to socioeconomic class. The lowest awareness percentage, around 40%, was observed in the poorest quintile (Q1), however this prevalence progressively climbed throughout categories, reaching over 80% in the wealthiest quintile (Q5). The gap in awareness between the highest and lowest wealth quintiles is significant, exceeding a two-fold difference.

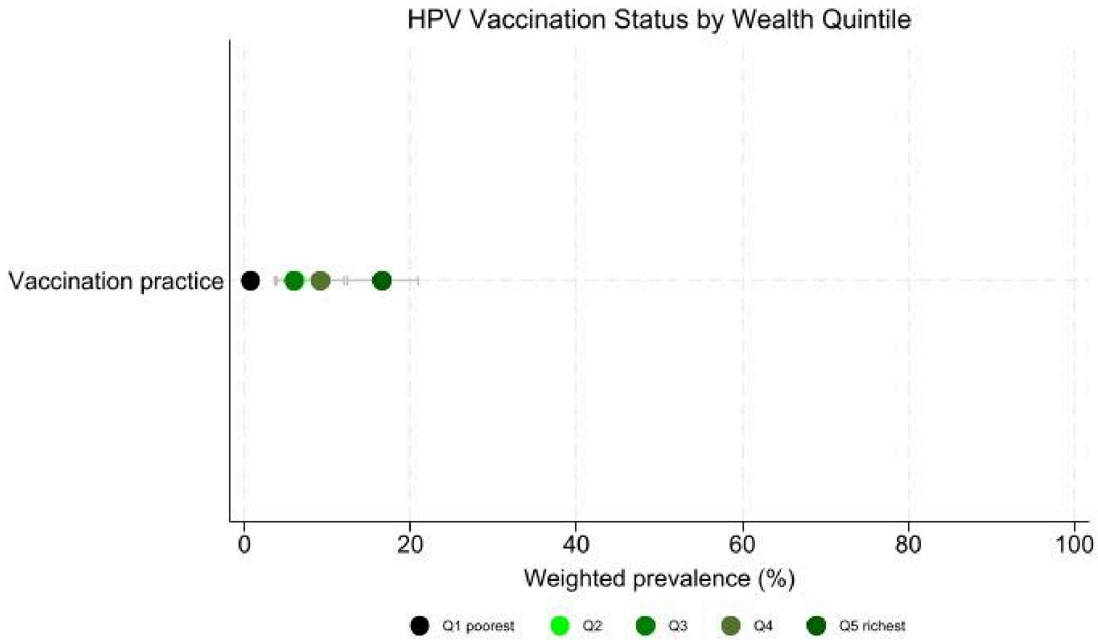


Figure 8. Status of the HPV vaccine and vaccination practices categorized by wealth quintile.

Economic inequality is profoundly evident in vaccination behavior. The poorest quintile (Q1) recorded a vaccination rate of nearly 0%. While this prevalence increased gradually across the Q2, Q3, and Q4 groups, a sharp upward surge was observed in the wealthiest quintile (Q5), reaching nearly 20%. This underscores that HPV vaccines currently remain accessible primarily to the highest socioeconomic segment.

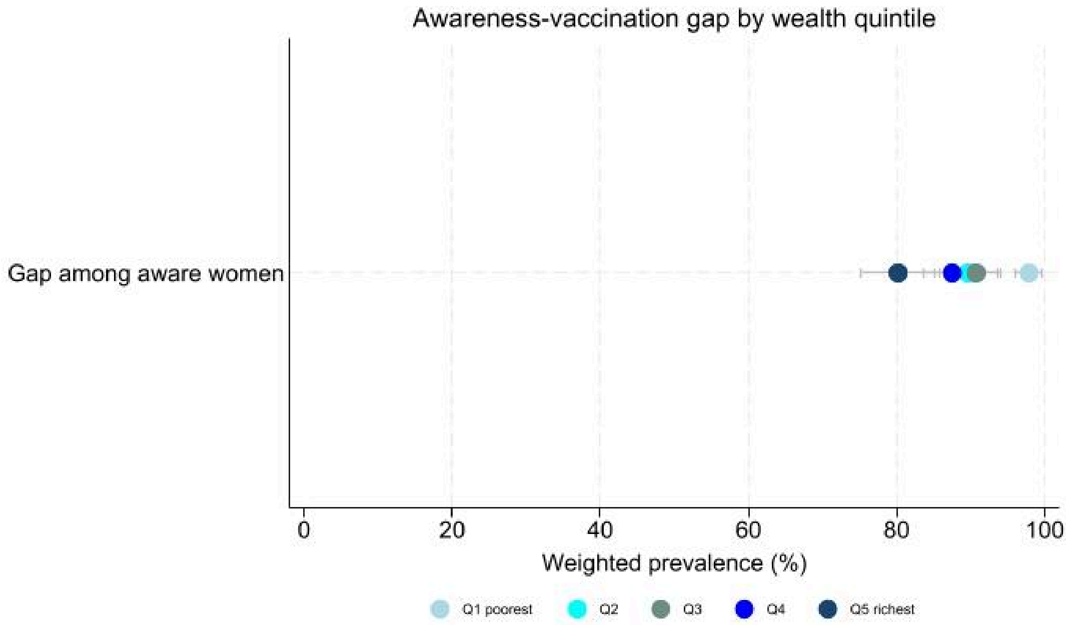


Figure 9. Knowledge-to-practice gap by wealth quintile (Gap among aware women)

Figure 9 illustrates a clear socioeconomic gradient in HPV vaccine awareness and uptake. Both awareness and vaccination increased with household wealth, but the gradient was more pronounced for vaccination uptake. Among women who were already aware of HPV vaccination, the awareness–vaccination gap was larger in poorer wealth quintiles, indicating that awareness alone may be insufficient to ensure vaccine uptake. This pattern is consistent with potential financial, service-access, and information-quality barriers under a predominantly private-pay vaccination context; however, the cross-sectional design does not allow confirmation of the specific mechanisms underlying this gap.

3.4.5 Decomposition of awareness inequality

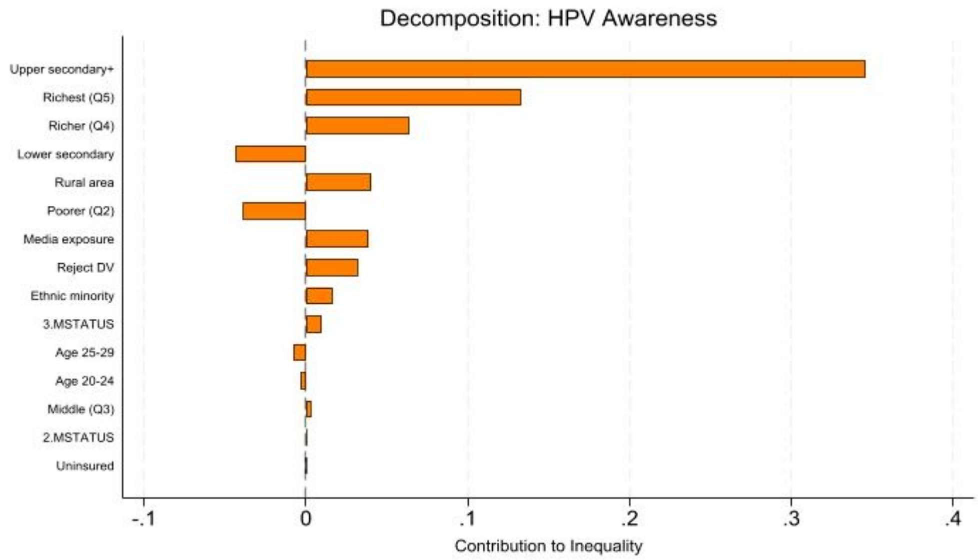


Figure 10. Decomposition analysis of socioeconomic inequality in HPV vaccine awareness.

Variables	dy/dx	CI	Contribution	% of total
Upper secondary and tertiary	0.2734	0.4348	0.3457	96.7244
Residual	-	-	-0.2311	-64.6408
Q5 (richest)	0.3257	0.5797	0.1328	37.1412
Q4	0.2202	0.3597	0.0637	17.8167
Lower secondary	0.1629	-0.3027	-0.0435	l-12.1826
Rural	-0.0418	-0.3849	0.0403	11.2720
Q2	0.1338	-0.3422	-0.0388	-10.8536
MME Frequent access	0.0633	0.1683	0.0387	10.8338
Rejects all wife-beating	0.0988	0.0938	0.0319	8.9225

Ethnic minority	-0.0716	-0.3792	0.0163	4.5660
Never married/in union	0.0218	0.2188	0.0096	2.6809
25-29	0.2763	-0.0181	-0.0080	-2.2342
20-24	0.2162	-0.0134	-0.0034	-0.9591
Q3	0.1437	0.0238	0.0030	0.8400
Formerly married/in union	-0.0809	-0.0285	0.0003	0.0856
Uninsured	0.0013	-0.0643	-0.0000	-0.0129

****In the decomposition analysis, a positive contribution percentage indicates that the variable exacerbates the pro-rich inequality. Conversely, a negative percentage indicates that the variable has an equalizing effect, acting to reduce the overall pro-rich inequality*

Table 9: Decomposition analysis of socioeconomic inequality in HPV vaccine awareness.

The decomposition analysis reveals that educational attainment and socioeconomic level are the principal factors contributing to disparities in HPV vaccination awareness. The most significant contributor to total inequality, accounting for 96.7% (Contribution = 0.3457), is possessing an upper secondary education or higher, indicating that educational background is the primary determinant of awareness levels. Wealth significantly influenced the disparity, with the wealthiest (Q5) and upper-middle (Q4) quintiles representing 37.1% and 17.8% of the difference, respectively. Conversely, variables such as rural domicile, ethnic minority status, and regular media exposure yielded comparatively little impacts, ranging from 4.5% to 11.2%.

3.4.6 Decomposition of vaccination inequality

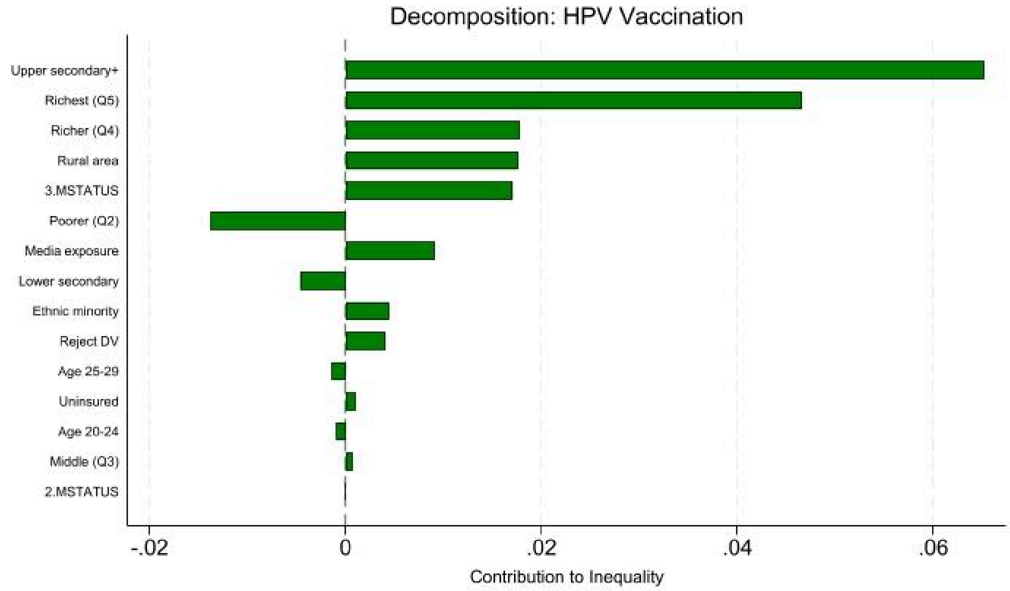


Figure 11. Decomposition analysis of socioeconomic inequality in HPV vaccination practice.

Variables	dy/dx	CI	Contribution	% of total
Upper secondary and tertiary	0.0515	0.4348	0.0652	61.9078
Residual	-	-	-0.0574	-54.4663
Q5 (richest)	0.1143	0.5797	0.0466	44.2400
Q4	0.0613	0.3597	0.0177	16.8429
Rural	-0.0182	-0.3849	0.0176	16.7002
Never married/in union	0.0388	0.2188	0.0171	16.2057
Q2	0.0475	-0.3422	-0.0138	-13.0764
MME Frequent access	0.0149	0.1683	0.0091	8.6412

Lower secondary	0.0173	-0.3027	-0.0046	-4.4034
Ethnic minority	-0.0195	-0.3792	0.0045	4.2302
Rejects all wife-beating	0.0125	0.0938	0.0040	3.8314
25-29	0.0481	-0.0181	-0.0014	-1.3215
Uninsured	-0.0289	-0.0643	0.0010	0.9672
20-24	0.0590	-0.0134	-0.0009	-0.8879
Q3	0.0344	0.0238	0.0007	0.6833
Formerly married/in union	0.0263	-0.0285	-0.0001	-0.0944

****In the decomposition analysis, a positive contribution percentage indicates that the variable exacerbates the pro-rich inequality. Conversely, a negative percentage indicates that the variable has an equalizing effect, acting to reduce the overall pro-rich inequality*

Table 10: Decomposition analysis of socioeconomic inequality in HPV vaccination practice.

Regarding vaccination uptake, the observed inequality remains heavily dictated by education and wealth, though social and demographic factors exert increased influence compared to awareness. Higher education continues to be the leading contributor at 61.9%, followed closely by the wealthiest quintile at 44.2%. Notably, never-married status and rural residency emerged as significant contributors, each accounting for approximately 16% of the overall inequality. These findings suggest that barriers to actual immunization are more multifaceted than those for awareness, involving a combination of significant economic burdens and specific demographic characteristics.

3.4.7 Decomposition of awareness–vaccination gap inequality

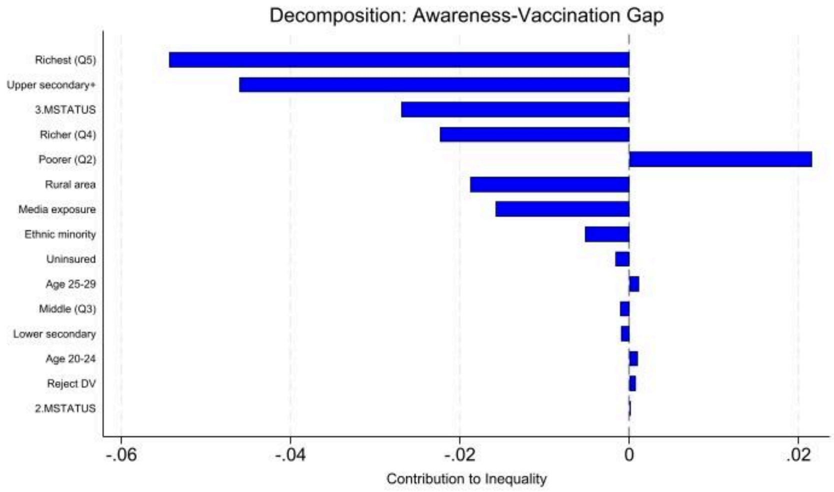


Figure 12. Decomposition analysis of socioeconomic inequality in the awareness-vaccination gap.

Variables	dy/dx	CI	Contribution	% of total
Residual	-	-	0.0625	-58.8660
Q5 (richest)	-0.1334	0.5797	-0.0544	51.2356
Upper secondary and tertiary	-0.0364	0.4348	-0.0461	43.3999
Never married/in union	-0.0612	0.2188	-0.0269	25.3813
Q4	-0.0774	0.3597	-0.0224	21.1058
Q2	-0.0744	-0.3422	0.0216	-20.3348
Rural	0.0195	-0.3849	-0.0188	17.6812
MME Frequent access	-0.0258	0.1683	-0.0158	14.8628
Ethnic minority	0.0230	-0.3792	-0.0052	4.9455

Uninsured	0.0453	-0.0643	-0.0016	1.5035
25-29	-0.0387	-0.0181	0.0011	-1.0537
Q3	-0.0516	0.0238	-0.0011	1.0148
Lower secondary	0.0037	-0.3027	-0.0010	0.9342
20-24	-0.0611	-0.0134	0.0010	-0.9124
Rejects all wife-beating	0.0024	0.0938	0.0008	-0.7403
Formerly married/in union	-0.0441	-0.0285	0.0002	-0.1572

****For the awareness–vaccination gap, the total concentration index is negative, indicating that the gap is concentrated among lower socioeconomic groups. Therefore, a positive percentage contribution means that the variable contributes in the same direction as the overall pro-poor concentration of the gap, whereas a negative percentage contribution offsets this pattern. The sign of the raw contribution should be interpreted together with the sign of the total concentration index.*

Table 11: Decomposition analysis of socioeconomic inequality in the awareness-vaccination gap.

The decomposition of the awareness–vaccination gap showed that this gap was disproportionately concentrated among lower socioeconomic groups. The wealthiest quintile (Q5) made the largest contribution to the measured inequality in the gap (51.2%), followed by upper secondary or tertiary education (43.4%) and never-married status (25.4%). However, these positive percentage contributions should be interpreted in light of the negative total concentration index. For Q5 and higher education, the negative marginal effects indicate that women in more advantaged groups were less likely to remain unvaccinated after becoming aware of HPV vaccination. Because these advantaged groups are concentrated among the wealthier, their lower probability of having an awareness–vaccination gap contributes to the overall concentration of the gap among poorer women. Thus, the decomposition does not imply that the wealthiest women experienced the largest gap; rather, it indicates that socioeconomic advantage was associated with a lower probability of the gap, thereby widening the relative disadvantage among poorer women.

CHAPTER IV: DISCUSSION

4.1. Interpretation of Prevalence and Awareness Levels

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The present study provides an overview of socioeconomic and demographic patterns related to HPV vaccine awareness and uptake among Vietnamese women aged 15–29 years. The findings suggest a substantial awareness–vaccination gap. Although 62.4% of women reported awareness of HPV vaccination, only 7.5% reported having received the vaccine. This indicates that awareness alone may be insufficient to ensure vaccination uptake. Vietnam's low uptake is consistent with patterns reported in several LMIC settings where HPV vaccination has not yet been fully integrated into national immunization programs. The socioeconomic differences observed in this study highlight important equity considerations for future public health policy.

4.2. Analysis of Associated Factors

Our analysis offers a novel "missing catch-up" group. Vietnam exhibits the lowest uptake (5.6%) in the 15-19 age range compared to High-Income Countries (HICs) such as Australia and the United Kingdom, where state-funded campaigns achieve more than 80% coverage in teens. The peak at ages 20-24 (10.0%) is indicative of delayed immunization until women reach financial independence, thereby missing the ideal biological window for maximum efficacy. The findings suggest that adolescents aged 15–19 may represent an important underserved group for future catch-up vaccination strategies. In this study, this age group had the lowest reported vaccination uptake. The higher uptake observed among women aged 20–24 may be consistent with delayed vaccination beyond early adolescence, although the cross-sectional design does not allow us to determine whether this pattern reflects financial independence, parental decision-making, service access, or other unmeasured factors. Moreover, we found a clear “pre-sexual” paradigm: unmarried women were 1.84 times more likely to get vaccinated than married women. In contrast, in China, sexual interaction often triggers immunization due to increased perceived vulnerability (You et al., 2020). In Vietnam, the increased uptake among the never-married reflects parental adherence to medical advice to vaccinate before sexual debut, indicating that family-led intervention is currently a strength, although restricted to those who can afford it. Furthermore, ethnic marginalization remains a severe barrier. The Kinh/Hoa majority (67.0%) had nearly double the level of knowledge as ethnic minorities (36.1%), while minority uptake was a paltry 1.6%. Substantial ethnic differences were also observed. The Kinh/Hoa majority had higher awareness and uptake than ethnic minority women, while vaccination uptake among ethnic minority women remained very low. This pattern may reflect multiple barriers faced by ethnic minority populations, including language, geographic access, cultural acceptability, and service availability. However, these mechanisms were not directly measured in the MICS data and should be interpreted cautiously. This ‘double burden’ of linguistic and geographic isolation is a global occurrence; even in HICs where free access is available, migrant and minority populations regularly experience worse uptake due to cultural mistrust (Dema et al., 2024).

4.3. The "Wealth Gap" in HPV Vaccination

In the Erreygers decomposition, upper secondary or higher education accounted for the largest share of the measured socioeconomic inequality in awareness (96.7%) .The contribution of education in this study appears larger than estimates reported in some studies from India and SSA settings (9.17% India and 6.23% SSA countries). In Vietnam, the strong association between education and awareness suggests that formal education may be an important pathway through which women access HPV-related information. This may reflect the interrelated roles of education, health literacy, vaccine knowledge, and trust in shaping awareness, although these pathways were not directly tested in this study 3. While education was strongly associated with awareness, household wealth showed a particularly strong association with vaccination uptake. In the unadjusted analysis, women in the richest quintile had substantially higher odds of vaccination than those in the poorest quintile. In the adjusted model, the wealth gradient remained strong. This pro-rich gradient is consistent with patterns often observed in settings where HPV vaccination is mainly available through fee-for-service or private-pay mechanisms. Similar socioeconomic differences in HPV vaccination or cervical cancer prevention have also been reported in some HICs, including the UK, USA, and Italy. Socioeconomic disadvantage remains strongly associated with unequal access to preventive health services across many health-system contexts. In the UK, cervical cancer incidence has been reported to remain higher in more deprived communities despite the presence of school-based prevention programs. In Vietnam, the findings suggest that information access alone may be insufficient to ensure vaccination uptake. Affordability, service availability, misinformation, and stigma may all contribute to the observed awareness–vaccination gap, although these mechanisms require further investigation.

4.4. Policy Implications

The expected rollout of HPV vaccination into Vietnam's EPI by the end of 2026 represents an important opportunity to reduce inequities in vaccine access. However, international evidence suggests that providing vaccination free of charge may not be sufficient on its own to eliminate inequalities in uptake. Future strategies should combine vaccine availability with targeted communication, culturally and linguistically appropriate messaging, and additional support for disadvantaged groups.

Specific catch-up strategies for women aged 15–19 may be needed to reduce the risk that this group is underserved during the EPI transition. Vietnam could also consider gender-neutral communication to frame HPV vaccination as a broader cancer prevention measure rather than an intervention linked only to female sexual behavior. In addition, targeted subsidies or bridging programs could be considered to improve access among women who may not benefit directly from routine school-based vaccination delivery. Because the study population aged 15–29 differs from the expected routine EPI target group, the findings should be interpreted as evidence of existing inequities among older adolescents and young adult women under the previous private-pay vaccination context, rather than as direct evidence of future EPI performance.

4.5. Strengths and Limitations.

4.5.1 Strengths

This study benefits from several robust methodological and contextual strengths that elevate the validity and reliability of its findings:

National Representativeness and Complex Survey Design

The study utilized the 2020-2021 Viet Nam Multiple Indicator Cluster Survey (MICS6), a high-quality, nationally representative dataset. The survey employed a rigorous two-stage, stratified cluster sampling design based on the 2019 Population and Housing Census. The sampling domains included urban and rural areas, six socio-economic regions, and the two major cities of Ha Noi and Ho Chi Minh City. Within these domains, enumeration areas were further stratified by the high and low proportions of ethnic minorities, resulting in the selection of 700 sample clusters nationwide. Because the MICS6 sample is not self-weighting, the application of precise sample weights (svy in Stata) was strictly adhered to in all statistical models. This comprehensive adjustment for stratification, clustering, and unequal selection probabilities ensures that the estimated prevalences and regression parameters are highly generalizable to the entire population of Vietnamese women aged 15–29.

Robust Measurement of Socioeconomic Status (Wealth Index)

Rather than relying on self-reported income, which is notoriously prone to reporting bias in low- and middle-income countries (LMICs), this study utilized the MICS6 wealth index. As detailed in the MICS6 report, this composite indicator was constructed using Principal Components Analysis (PCA) based on household ownership of consumer goods, dwelling characteristics, water and sanitation facilities, and other asset-related indicators. To minimize urban bias—a common pitfall in asset-based indices—separate factor scores were initially calculated for urban and rural households before being regressed onto the initial total sample scores to create a harmonized, national wealth ranking. This sophisticated methodology provided a highly stable and accurate reflection of long-term household purchasing power, enabling a precise application of the Concentration Index and Erreygers decomposition.

Exploration of Nuanced Sociocultural Factors (Gender Equity and Media Access)

The MICS6 dataset provides a distinctive advantage by facilitating the analysis of nuanced sociocultural determinants of health behaviour, specifically in relation to gender equity and media access, beyond traditional demographic variables. Mass Media and ICT Exposure (MME): The MME variable accurately measured the frequency of interaction with information channels, defining regular exposure as engaging with a newspaper, radio, or television at least once a week. The analysis of MME in this study enabled an assessment of the impact of multidimensional access to health information on the "knowledge-to-action" gap in HPV prevention, pinpointing actionable priorities for future digital and mass media health campaigns. The study assessed gender-equitable attitudes by examining female empowerment using a proxy variable that ascertains if a woman rejects all justifications for domestic violence (e.g., marital disagreements, child neglect, or denial of sexual connections). Incorporating this variable into the model provided substantial insights into the relationship between female autonomy and entrenched patriarchal norms about healthcare-seeking habits. The choice to get a pricey preventive measure like the HPV vaccine is not just about money; it also has a lot to do with a woman's freedom in her own home.

Methodological Rigor in Inequality Analysis

The use of Directed Acyclic Graphs (DAGs) established a solid theoretical framework for the identification of confounders, hence reducing the likelihood of over-adjustment or collider bias. Additionally, employing the Erreygers-corrected Concentration Index directly tackled the boundary problems associated with binary health outcomes (awareness and vaccination), guaranteeing that the decomposition results were mathematically valid and comparable among various population segments.

4.5.2 Limitations

Despite its robust methodology, the study acknowledges several limitations that should be considered when interpreting the findings:

Cross-Sectional Design

Due to the cross-sectional nature of the MICS6 dataset, the study can only establish associations rather than true causal relationships between socioeconomic determinants and HPV-related outcomes.

Self-Reporting Bias

The data regarding HPV vaccination status and awareness were self-reported, potentially introducing recall bias or social desirability bias. The inability to cross-reference replies with clinical or national immunisation registries poses a risk of inaccurately estimating genuine vaccination coverage.

Sociocultural Confounding in Sexual Behavior Reporting (Multicollinearity)

A significant limitation encountered during modeling was the substantial multicollinearity (Variance Inflation Factor > 11) between self-reported sexual behavior and marital status, especially among the 'never married/in union' group. In traditional Vietnamese culture, which is heavily influenced by Confucian values, premarital sex is still very much frowned upon. Consequently, unmarried young women typically do not reliably report sexual behavior in face-to-face cross-sectional surveys. Because the self-reported sexual debut and marital status in this dataset were almost perfectly correlated, the sexual behavior variable was left out of the multivariable models to keep the statistics accurate. This absence, albeit essential, limits the study's ability to pinpoint the specific, risk-based motivations for HPV vaccination among unmarried, sexually active women.

Unobserved Confounding and Large Residuals in Decomposition

A notable limitation in the Erreygers decomposition analysis is the substantial size of the remaining components (e.g., -64.6% for awareness and -54.4% for immunisation). When you use linear decomposition methods on non-linear models (like Probit), you often get big residuals because of approximation errors. These big residuals also show that there are confounding factors that you can't see. The MICS6 dataset did not include important structural and health-system characteristics, such as how close private vaccination clinics (like VNVC centers) are to each other, how vaccine supply chains differ from region to region, or how local health communication campaigns work. These unobserved determinants probably have a big impact on the inequality landscape, which means that the real causes of the HPV vaccination gap go beyond individual wealth and education.

CONCLUSION AND RECOMMENDATIONS

The results of this study indicate substantial socioeconomic inequality in HPV vaccine awareness and uptake among Vietnamese women aged 15–29 years. Although 62.4% of women reported having heard of HPV vaccination, only 7.5% reported having received the vaccine, suggesting a considerable awareness–vaccination gap. This pattern indicates that awareness alone may be insufficient to ensure vaccine uptake and that other socioeconomic and structural factors may influence whether women are able to translate knowledge into preventive action.

Data from the MICS 6 survey showed that educational attainment was strongly associated with HPV vaccine awareness. In the decomposition analysis, upper secondary or higher education accounted for the largest share of the measured socioeconomic inequality in awareness. This suggests that formal education may be an important pathway through which women access HPV-related information. However, education should not be interpreted as the only source of HPV knowledge, as awareness may also be influenced by media exposure, healthcare contact, family networks, and community-level communication. Women with lower educational attainment may therefore face greater barriers to accessing accurate and timely information about HPV vaccination.

Household wealth was also strongly associated with HPV vaccination uptake. Women in the highest wealth quintile had substantially higher odds of reporting HPV vaccination than those in the lowest quintile. This pro-rich pattern suggests that, under a predominantly fee-for-service vaccination model, affordability and service access may have contributed to unequal uptake. However, because the study used cross-sectional data, these findings should be interpreted as associations rather than evidence of direct causation. The very low vaccination uptake among ethnic minority women further highlights the need to consider overlapping disadvantages related to socioeconomic status, education, ethnicity, and access to health services.

The awareness–vaccination gap was concentrated among lower socioeconomic groups, suggesting that poorer women were less able to translate vaccine awareness into actual uptake. Decomposition results showed that wealth and education were the largest contributors to this inequality pattern. This does not imply that advantaged groups experienced a larger gap. Rather, because the total concentration index for the gap was negative, the results indicate that women in more advantaged socioeconomic groups were less likely to remain unvaccinated after becoming aware of HPV vaccination, thereby making the gap more concentrated among poorer women.

Furthermore, the reduced role of urban–rural residence in the multivariable models suggests that socioeconomic and educational differences may be more important than physical location alone in explaining HPV vaccination inequalities. The higher uptake among never-married women may reflect vaccination decisions occurring before marriage or sexual debut, potentially influenced by family guidance, healthcare advice, or perceived need. However, these mechanisms were not directly measured in the MICS data and should therefore be interpreted cautiously.

In conclusion, efforts to eliminate cervical cancer in Vietnam will require more than awareness-raising campaigns. The planned integration of HPV vaccination into Vietnam's Expanded Program on Immunization represents an important opportunity to reduce financial barriers and improve equitable access. Nevertheless, free vaccine provision alone may not be sufficient.

Future programs should combine vaccine availability with targeted communication strategies, culturally and linguistically appropriate outreach, and additional support for disadvantaged groups, including poorer women, women with lower educational attainment, ethnic minority women, and those who may not be reached through school-based delivery. These findings provide evidence to inform equity-oriented HPV vaccination policy during Vietnam’s transition toward national program implementation. Ultimately, reducing socioeconomic inequalities in cervical cancer prevention will require addressing both affordability and unequal access to reliable vaccine information.

REFERENCES

APPENDIX

Table : Summary of explanatory variables used in the analysis

Variables	Categories	Operational Definition	Note
Age group (WAGE)	15-19, 20-24, 25-29	MICS standard 5-year age group	
Residence (HH6)	Urban, Rural	MICS household classification	
Marital status (MSTATUS)	Currently married/union, Formerly married, Never married	MICS marital status variable	
Education (edu)	Primary or less, Lower secondary, Upper secondary+	Collapsed from MICS welevel	
Wealth quintile (ses5)	Q1(poorest) to Q5(richest)	MICS wealth index quintiles	
Health insurance (insurance)	No, Yes	MICS insurance variable; coded as uninsured = 1, insured = 0	
Mass media & ICT exposure	No, Yes	Any exposure to newspaper, radio, or TV	MT1 (Newspaper), MT2 (Radio), MT3 (TV) MT5 (Computer), MT12 (Phone), MT10 (Internet)

Ethnicity (minority)	Kinh/ Hoa (No), Ethnic minority (Yes)	MICS variable ethnicity2	
Rejection of wife-beating (reject wb all)	Does not reject, Rejects all	Rejects wife-beating in all 5 DV1 scenarios	(DV1A–DV1E) proxy for women's empowerment
Sexual experience (sb1n)	No, Yes	Ever had sexual intercourse (SB1)	

Table : Variance Inflation Factor (VIF) for independent variables (Appendix)

Variables	**VIF**	**VIF (except sex behavior)**
Age group		
20-24	1.69	1.66
25-29	2.32	2.27
Residence		
Rural	1.33	1.33
Ethnicity		
Ethnic minority	2.37	2.37

Marital status		
Formerly married/union	1.03	1.03
Never married/union	11.51	2.02
Education		
Lower secondary	2.37	2.37
Upper secondary and tertiary	3.44	3.44
Wealth quintile		
Q2	1.79	1.79
Q3	2.09	2.09
Q4	2.24	2.24
Q5 (richest)	2.36	2.36
Health insurance		
Uninsured	1.05	1.04
Mass media & ICT exposure		
Frequent access	1.39	1.39
Gender-equitable attitude		
Rejects all wife-beating	1.06	1.05
Sexual behavior		

Yes	11.44	-
Mean VIF	2.86	1.79